Recommendation System

Introduction

Implicit（隱含式）Data：使用者活動如點擊的結連結、購買的產品 較容易蒐集

Explicit（明確式）Data：評分、留言

Collaborative Filtering：根據回應分類

Content Based：根據劇情內容分類

Alogrithm （尋找 implict data 的關聯性）

Association Rule Mining（關聯規則探勘 ARM）：使用Apriori（先驗演算）

Alternating Least Square（交替最小平方法 ALS）：collaborative filtering algorithm

Content Based Filtering（內容過濾）：使用物品特色讓使用者被推薦到

Association Minig (Support and Confidence)

Support（支持度） $\text{Support}(A \to C) = P(A \cap C)$

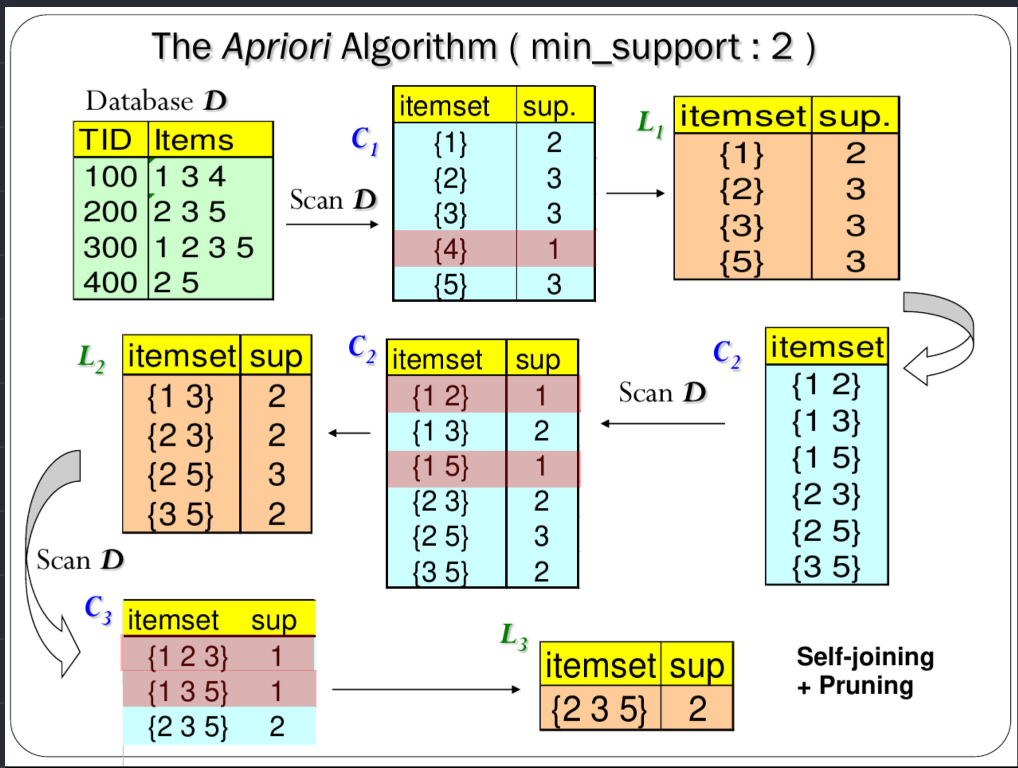
Confidence(信賴度) $\text{Confidence}(A \to C) = P(C|A) = P(A \cap C) \div P(A)$

Lift（增益率）： 規則多有效 $\text{Lift}(A \to C) = P(C|A) \div P(C) = P(A \cap C) \div [P(A) \times P(C)]$

Apriori algorithm

Transaction ID	Items Bought
2000	A,B,C
1000	A,C
4000	A,D
5000	B,E,F

Frequent Itemset	Support
{A}	75%
{B}	50%
{C}	50%
{A,C}	50%



Self-joining and Pruning:

$L_3 = \{abc, abd, acd, ace, bcd\}$

Self-Joining: $L_3 \cdot L_3 = \{abcd, abce, acde\}$

Pruning: abce is removed because abe is not in L3

acde is removed because ade is not in L3

Performance

Bad performance due to scanning database n+1 times (n is the longest pattern)

Frequent Patterns

Benifit

1. No candidate generation
2. Only need to scan database 2 times
3. Completeness (完整) and Compactness (緊密) in FP-Tree

Steps

1. Scan DB once, find frequent 1-itemset (single item pattern)
2. Order frequent items in frequency descending order
3. Scan DB again, construct FP-tree

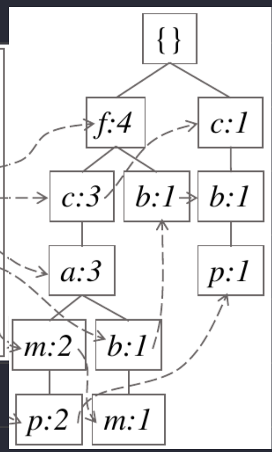
Example

Min-Support = 50% (count ≥ 3)

TID	*Items bought*	*(ordered) frequent items*
100	{*f, a, c, d, g, i, m, p*}	{*f, c, a, m, p*}
200	{*a, b, c, f, l, m, o*}	{*f, c, a, b, m*}
300	{*b, f, h, j, o*}	{*f, b*}
400	{*b, c, k, s, p*}	{*c, b, p*}
500	{*a, f, c, e, l, p, m, n*}	{*f, c, a, m, p*}

Header Table

Item	*frequency*	*head*
f	*4*	
c	*4*	
a	*3*	
b	*3*	
m	*3*	
p	*3*	

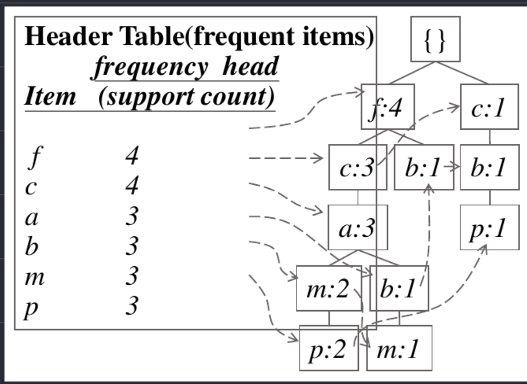


Mining Frequent Patterns Using FP-tree

1. Construct conditional pattern-base for each item
2. This will turn FP-Tree into conditional FP-Tree
3. Repeat the process on each newly created conditional FP-Tree
4. Stop until FP-tree is empty or it only contains 1 path

Example

Step 1 From FP-tree to Conditional Pattern Base

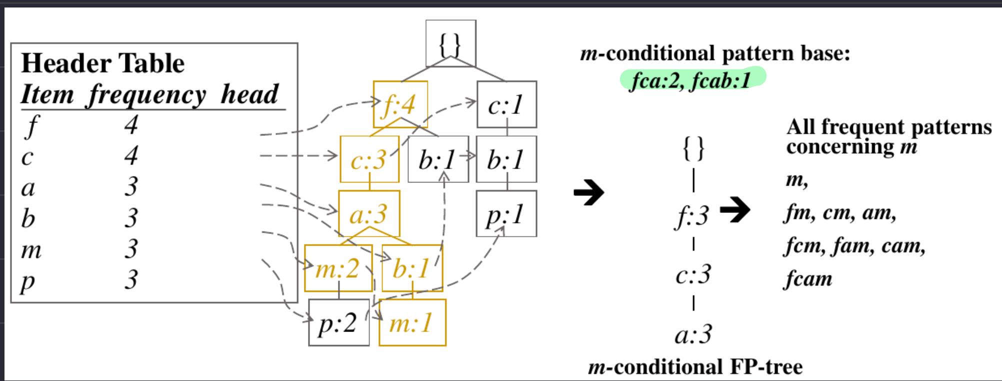


Conditional pattern bases

item	cond. pattern base
c	f:3
a	fc:3
b	fca:1, f:1, c:1
m	fca:2, fcab:1
p	fcam:2, cb:1

Step 2: Construct Conditional FP-tree for each children (min supp·=3)

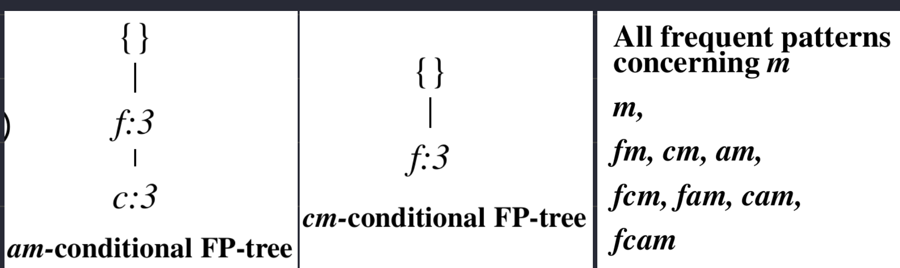
本張圖片僅展示 m-conditional base (從 m 向上) 的 FP-Tree 建置方式



雖然 b 是 m 的母節點，但是因為其不符合 min-support 的要求，所以可以跳過

Item	Conditional pattern-base	Conditional FP-tree
p	{(fcam:2), (cb:1)}	{(c:3)}|p
m	{(fca:2), (fcab:1)}	{(f:3, c:3, a:3)}|m
b	{(fca:1), (f:1), (c:1)}	Empty
a	{(fc:3)}	{(f:3, c:3)}|a
c	{(f:3)}	{(f:3)}|c
f	Empty	Empty

Step 3 Recursively mine the conditional FP-tree



Finding Similarity

User Collaborative Filtering

Jaccard Similarity

只觀察同樣的部分，完全不看兩者的數值大小關係

$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$

Cosine Similarity

將雙方資料當成向量作內積計算

$\text{sim}(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2} \cdot \sqrt{\sum_{i=1}^{n} y_i^2}}$

Pearson Correlation Coefficient

$r_{xy} = \frac{\sum_{i \in I_{xy}} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i \in I_{xy}} (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i \in I_{xy}} (y_i - \bar{y})^2}}$

Example

User	Item 1	Item 2	Item 3	Item 4	Item 5
u_1	⋆⋆⋆⋆⋆	⋆⋆⋆	−	⋆	⋆⋆⋆⋆
u_2	⋆⋆	−	⋆⋆⋆⋆⋆	⋆⋆⋆	⋆

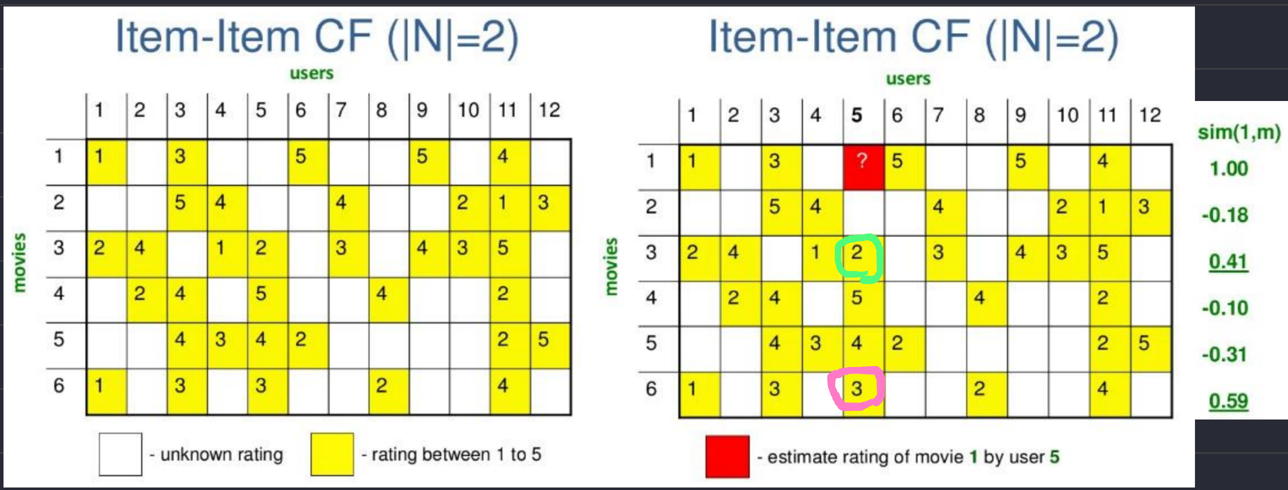
$r_1 = \{5, 3, 0, 1, 4\}$ $r_2 = \{2, 0, 5, 3, 1\}$

Jaccard: $J(u_1, u_2) = 3 \div 5$

Cosine: $\frac{r_1 \cdot r_2}{|r_1| \times |r_2|} = \frac{10 + 0 + 0 + 3 + 4}{\sqrt{51} \times \sqrt{39}} = \frac{17}{\sqrt{51} \times \sqrt{39}}$

Item-Based Collaborative Filtering

使用所有使用者給予的評價，比較物品之間的關聯性



Item-Item CF (|N|=2)

users / movies

	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3			5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

- unknown rating
- rating between 1 to 5

Item-Item CF (|N|=2)

users / movies

	1	2	3	4	5	6	7	8	9	10	11	12	sim(1,m)
1	1		3		?	5			5		4		1.00
2			5	4			4			2	1	3	-0.18
3	2	4		1	2		3		4	3	5		0.41
4		2	4		5			4			2		-0.10
5			4	3	4	2					2	5	-0.31
6	1		3		3			2			4		0.59

- estimate rating of movie 1 by user 5

Step 1: 找到與電影1相似的點影（假設是 3 與 6）

Step 2: 將 User 5 對 電影3 與 電影6 的評價讀取出來

Step 3: 帶入相似度進行加權平均 獲得 分數 $\hat{r}_{5,1} = \frac{(s_{1,3} \times r_{5,3}) + (s_{1,6} \times r_{5,6})}{s_{1,3} + s_{1,6}}$